

Tushar Kankhedia

Full Stack Developer | IIT Jodhpur Graduate

@ tusharkankhedia13@gmail.com

+91 7610020788

Hyderabad, India

LinkedIn

Experience

Nebulaa Innovations Pvt Ltd (ParadigmIT Technology Services Pvt Ltd)

System Engineer – Full Stack Developer (Graduate Trainee)

Dec 2023 – Present

Hyderabad

- Led the development of **scalable full-stack systems** using React and Flask, optimizing **backend workflows**, **PostgreSQL queries**, and **Redis caching** to improve **performance, reliability, and maintainability** across production environments.
- Cut release time by 90%**; owned and implemented a **CI/CD pipeline (Bitbucket Pipelines)** with automated testing and deployment
- Reduced average ticket resolution time by 50%** by redesigning and optimizing the **web-based support system** and streamlining workflow processes
- Developed a **real-time analytics dashboard** enabling centralized KPI monitoring and efficient handling of **millions of records** through optimized query indexing, partitioning, and data aggregation workflows.
- Independently designed and built a custom ERP system from the ground up**, enhancing data management and significantly improving operational efficiency across departments; enabled transparent inventory tracking and delivered actionable analytics to ensure optimal stock availability.
- Implemented a scalable security framework using Role-Based Access Control (RBAC) and JWT authentication**, enhancing data privacy, securing endpoints, and enabling fine-grained access control.
- Tech Stack:** React.js, JavaScript, Flask, PostgreSQL, Redis, Bitbucket Pipelines, Material-UI, Bootstrap, Recharts

Projects

Alex AI Assistant

Mar 2026 – Present

- Engineered a **RAG eval harness** with 55 labeled benchmark queries — measuring retrieval precision (0.66), context recall (0.74), faithfulness (0.73), and section hit rate (96.4%) — with CLI-driven experiment tracking for repeatable before/after comparisons across embedding models, chunking strategies, and retrieval configs
- Implemented advanced retrieval pipeline — **semantic chunking, multi-query retrieval, reranking** — with **Redis semantic caching** reducing LLM API calls by **18.2%** across 55 benchmark queries
- Migrated vector store from **local FAISS to Pinecone** for persistent cloud-native retrieval; integrated **Cloudflare Workers AI** to decouple edge-based embedding generation from core LLM inference costs
- Shipped to production: **FastAPI on Render + React on Vercel** — full ownership from RAG architecture to live deployment
- Tech Stack:** Python, FastAPI, LangChain, LangSmith, Pinecone, Redis, Cloudflare Workers AI, Google Gemini, AsyncIO, React.js, Render, Vercel

Co-curricular Activities

- 2021 Student Volunteer in Convocation Events Management Team
- 2019 Student Volunteer in Ignus Publicity and Media Team

Technical Skills

- Languages:** Python, SQL, JavaScript
- AI Systems:** RAG, LangChain, LangSmith, Google AI APIs, Cloudflare Workers AI
- Retrieval:** Pinecone, FAISS, Semantic Chunking, Hybrid Search, Reranking
- Evaluation:** RAG Eval Harness, LLM Evaluation, Hallucination Tracking, Retrieval Benchmarking
- Backend:** FastAPI, Flask, AsyncIO
- Data:** PostgreSQL, Redis, Alembic
- Infra:** Docker, Nginx, Render, Vercel, Bitbucket Pipelines
- Frontend:** React.js, Material-UI, Bootstrap

Achievements

- HackerRank Gold Badge (Python)**
- Solved 300+ DSA problems**
- Python Certificate (HackerRank)**
- Problem Solving Certificate (HackerRank)**

Education

Bachelor of Technology (B.Tech) in Mechanical Engineering

Indian Institute of Technology, Jodhpur

2019 – 2023

Course Work

- Python for Everybody
- Python for Data Science, AI & Development
- Data Structures and Algorithms
- Introduction to Computer Science
- Introduction to Machine Learning
- Introduction to IoT Systems
- Computer Networks (CN)

Links

- GitHub:** View Profile
- GeeksforGeeks:** View Profile
- HackerRank:** View Profile